



NGS SEQUENCING ANALYSIS

RNA-SEQ: PART 2

```
object to mirror_mod.mirror_object
operation == "MIRROR_X":
mirror_mod.use_x = True
mirror_mod.use_y = False
mirror_mod.use_z = False
operation == "MIRROR_Y":
mirror_mod.use_x = False
mirror_mod.use_y = True
mirror_mod.use_z = False
operation == "MIRROR_Z":
mirror_mod.use_x = False
mirror_mod.use_y = False
mirror_mod.use_z = True
```

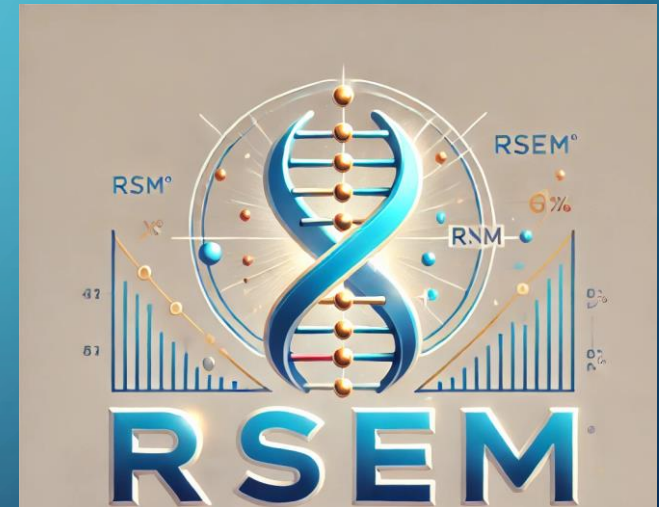
```
selection at the end - add
ob.select= 1
ier_ob.select=1
context.scene.objects.active
("Selected" + str(modifier
mirror_ob.select = 0
bpy.context.selected_ob
data.objects[one.name].se
print("please select exactl
```

--- OPERATOR CLASSES ---

```
types.Operator):
X mirror to the selected
object.mirror_mirror_x"
error X"
```

RECAP

- We performed quality control (QC) on the sequencing files using FastQC and MultiQC
- We aligned the sequencing reads to the reference genome using STAR aligner
- We quantified transcripts using RSEM



THE ALIGNMENT SCRIPT

```
#!/bin/bash

#SBATCH -J alignment
#SBATCH -o alignment.out
#SBATCH -N 1 -n 32

module load star
module load RSEM

mkdir reference/star_index
mkdir reference/rsem_index
mkdir results
mkdir results/final

gunzip reference/GRCh38.primary_assembly.genome.fa.gz
gunzip reference/gencode.v47.primary_assembly.basic.annotation.gtf.gz
```

THE ALIGNMENT SCRIPT

Creating indexes for STAR and RSEM

```
STAR --runMode genomeGenerate --runThreadN 32 --genomeDir reference/star_index --genomeFastaFiles reference/GRCh38.primary_assembly.genome.fa --sjdbGTFfile reference/gencode.v47.primary_assembly.basic.annotation.gtf  
rsem-prepare-reference --gtf reference/gencode.v47.primary_assembly.basic.annotation.gtf reference/GRCh38.primary_assembly.genome.fa reference/rsem_index/rsem_index
```

Using a for-loop to align the sequences to the reference genome (STAR) and quantify the transcripts (RSEM)

```
for p in SRR*; do  
    STAR --runThreadN 32 --genomeDir reference/star_index --readFilesIn "$p/${p}_1.fastq.gz" "$p/${p}_2.fastq.gz" --outFileNamePrefix results/"$p." --outSAMtype BAM SortedByCoordinate --quantMode TranscriptomeSAM GeneCounts  
    rsem-calculate-expression -p 32 --paired-end --alignments results/"$p".Aligned.toTranscriptome.out.bam reference/rsem_index/rsem_index results/"$p"  
    sed "s/\\.[0-9]*//" results/"$p".genes.results > results/final/"$p".genes.results.clean  
done
```

Compressing files and removing temporary folders

```
pigz -p 32 results/*.bam  
rm -r results/*.tmp  
rm -r results/*.stat  
rm -r results/*_STARTmp
```

TODAY

- Perform differential expression analysis on the results



OPEN ACCESS PEER-REVIEWED

RESEARCH ARTICLE

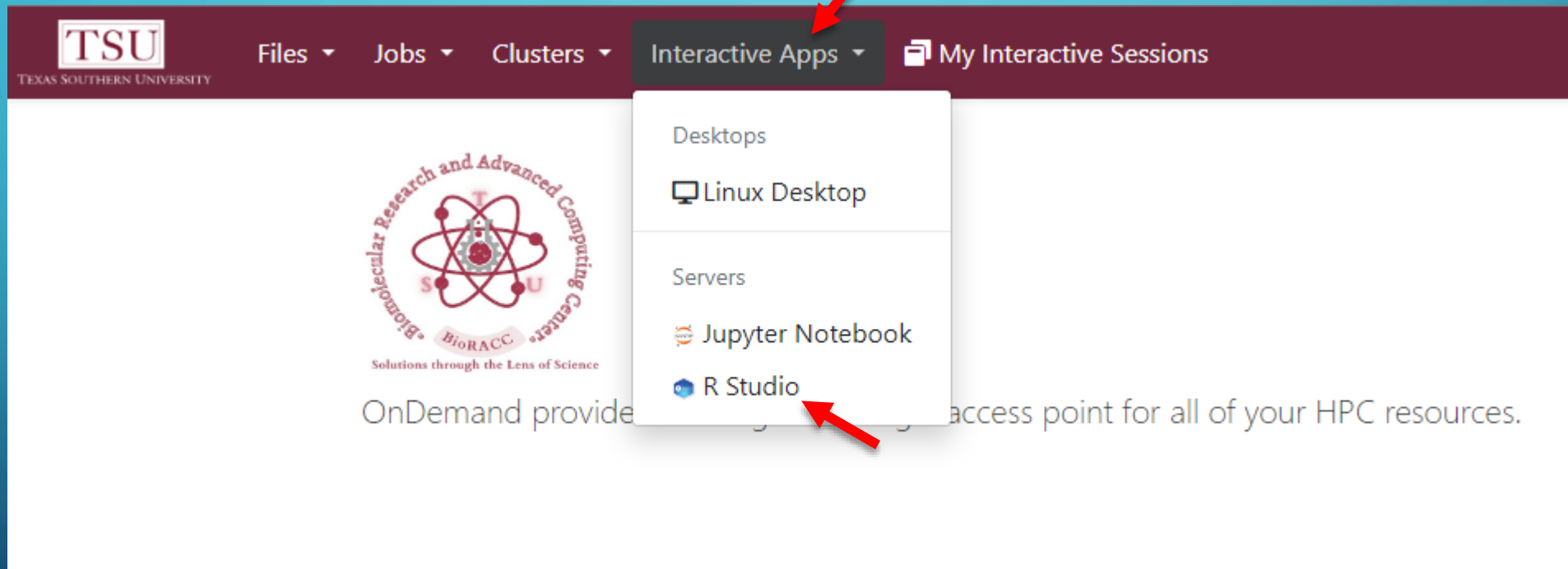
Transcriptomic analysis of cellular senescence induced by ectopic expression of ATF6 α in human breast cancer cells

Ju Won Kim , So-Hyun Bae , Yesol Moon, Eun Kyung Kim, Yongjin Kim, Yun Gyu Park, Mi-Ryung Han , Jeongwon Sohn

Published: October 28, 2024 • <https://doi.org/10.1371/journal.pone.0309749>

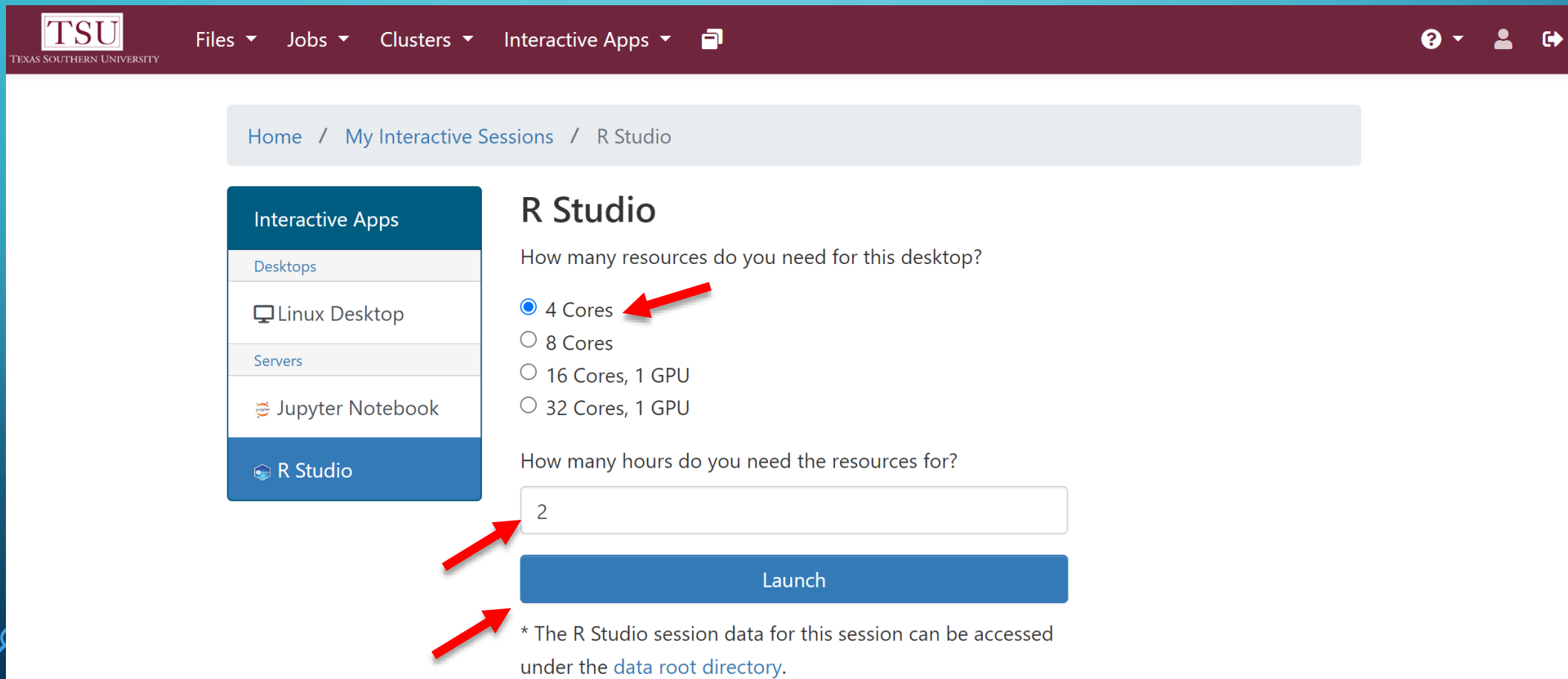
REQUESTING AN RSTUDIO SESSION

Interactive Apps → R Studio



REQUESTING AN RSTUDIO SESSION

Select **4 Cores** and enter **2** for the number of hours, then click Launch



TSU
TEXAS SOUTHERN UNIVERSITY

Files ▾ Jobs ▾ Clusters ▾ Interactive Apps ▾

Home / My Interactive Sessions / R Studio

Interactive Apps

- Desktops
- Linux Desktop
- Servers
- Jupyter Notebook
- R Studio**

R Studio

How many resources do you need for this desktop?

☒ 4 Cores

☐ 8 Cores

☐ 16 Cores, 1 GPU

☐ 32 Cores, 1 GPU

How many hours do you need the resources for?

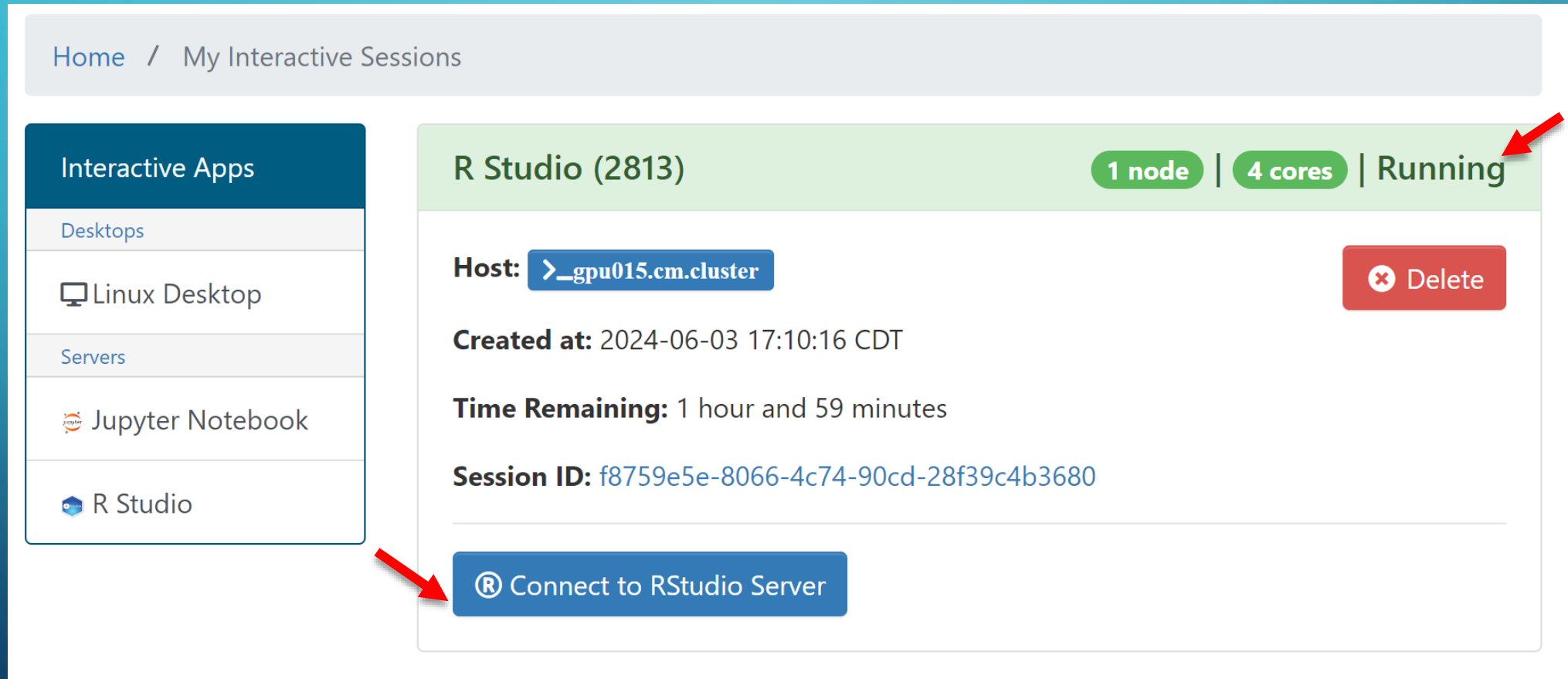
2

Launch

* The R Studio session data for this session can be accessed under the [data root directory](#).

REQUESTING AN RSTUDIO SESSION

Click *Connect to Rstudio Server* when session is running



The screenshot displays a web interface for managing interactive sessions. At the top, a breadcrumb trail shows 'Home / My Interactive Sessions'. On the left, a sidebar titled 'Interactive Apps' contains a list of options: 'Desktops', 'Linux Desktop', 'Servers', 'Jupyter Notebook', and 'R Studio'. The 'R Studio' option is selected. The main content area shows details for an 'R Studio (2813)' session. The session status is 'Running', with '1 node' and '4 cores' allocated. A red arrow points to the 'Running' status. Below the status, the 'Host' is '>_gpu015.cm.cluster', 'Created at' is '2024-06-03 17:10:16 CDT', 'Time Remaining' is '1 hour and 59 minutes', and 'Session ID' is 'f8759e5e-8066-4c74-90cd-28f39c4b3680'. A red 'Delete' button is located next to the host information. At the bottom of the session details, a blue button labeled '® Connect to RStudio Server' is highlighted with a red arrow.

Home / My Interactive Sessions

Interactive Apps

- Desktops
- Linux Desktop
- Servers
- Jupyter Notebook
- R Studio

R Studio (2813) 1 node | 4 cores | Running

Host: >_gpu015.cm.cluster ✕ Delete

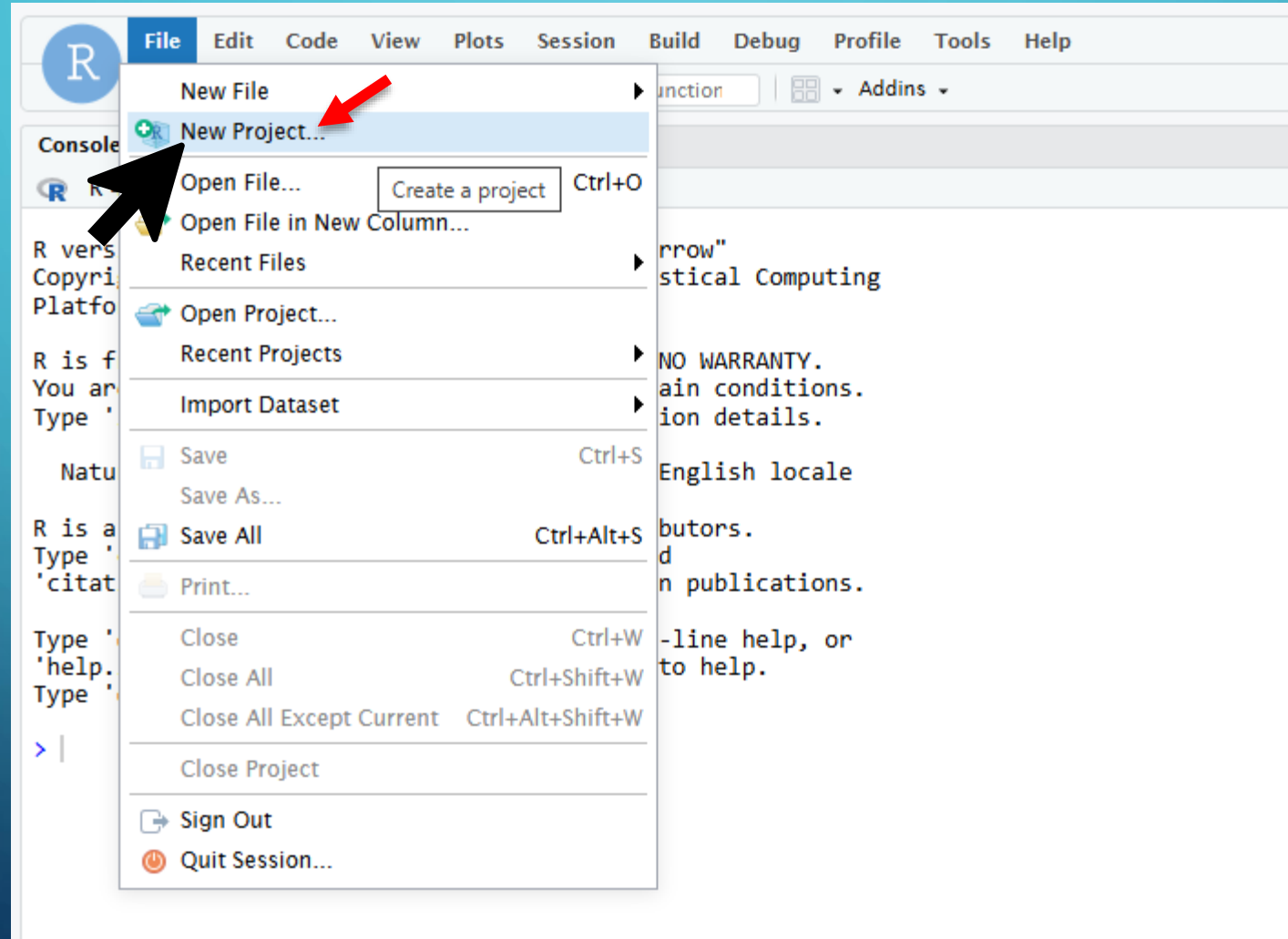
Created at: 2024-06-03 17:10:16 CDT

Time Remaining: 1 hour and 59 minutes

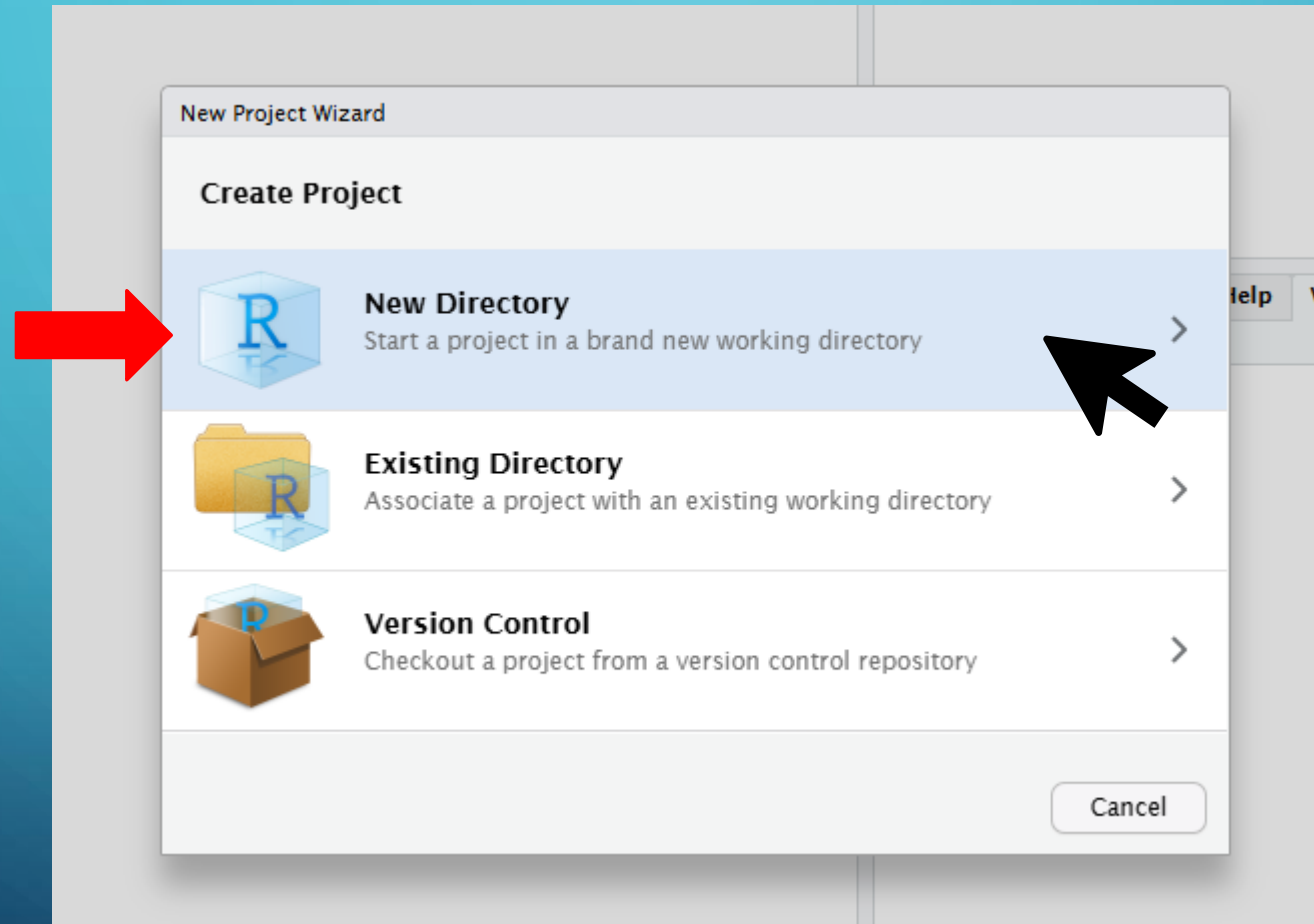
Session ID: f8759e5e-8066-4c74-90cd-28f39c4b3680

® Connect to RStudio Server

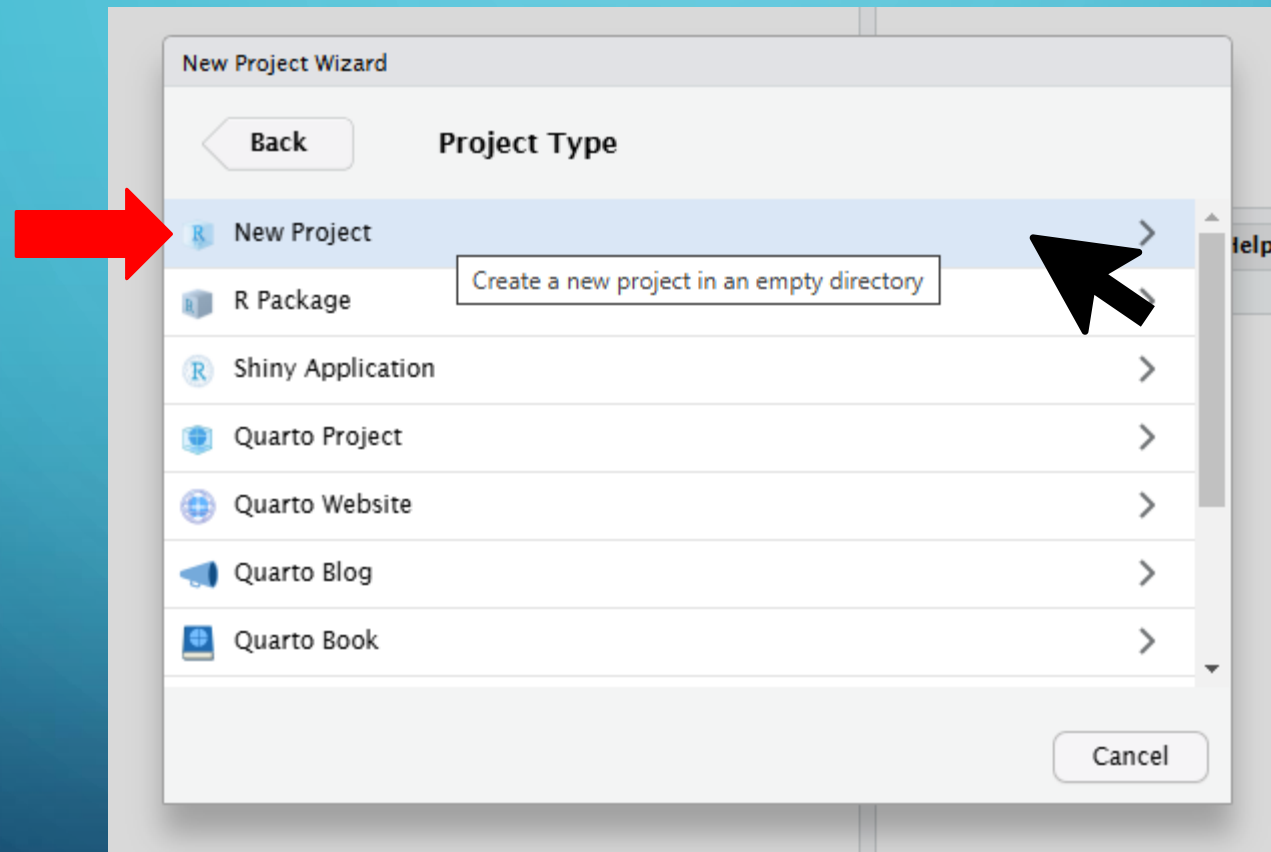
CREATING A PROJECT



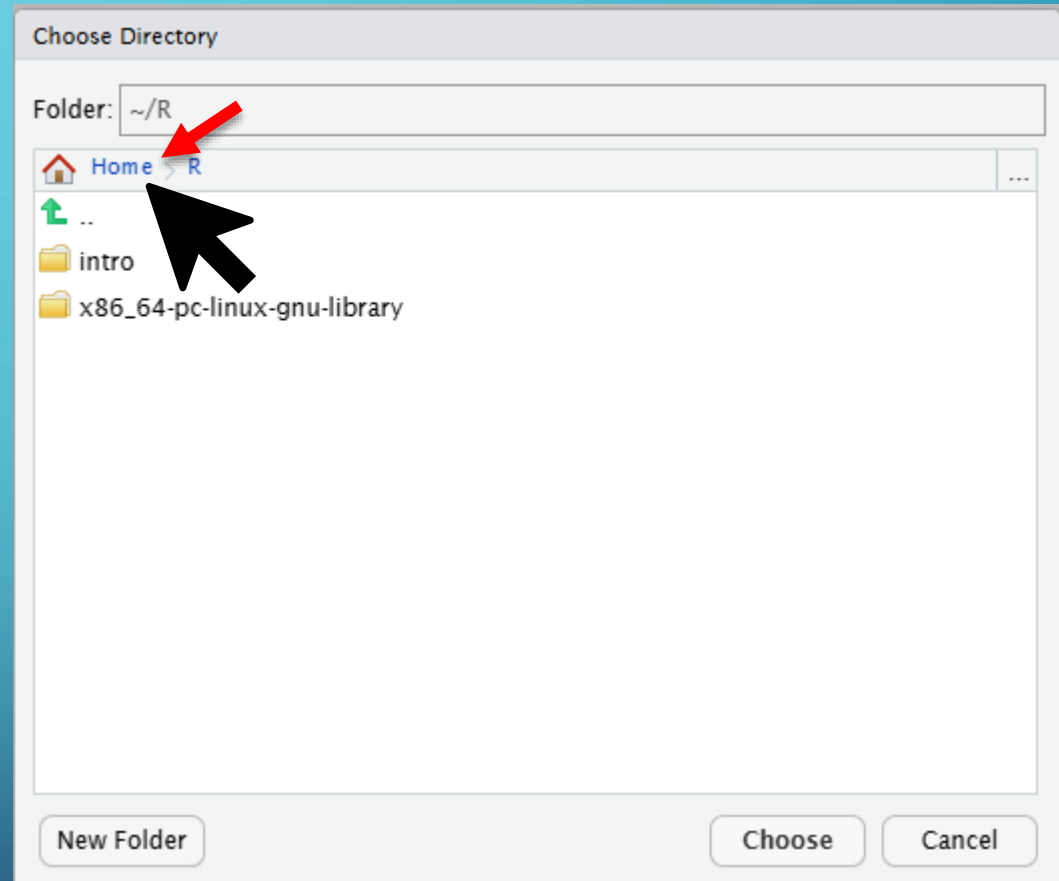
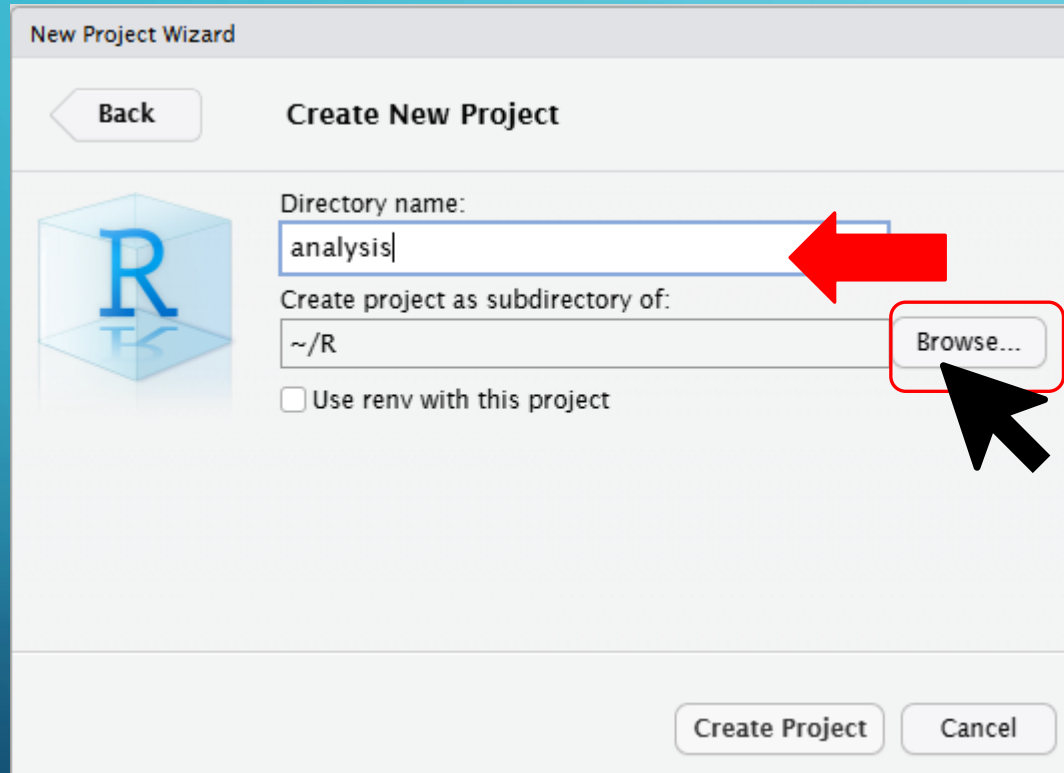
CREATING A PROJECT



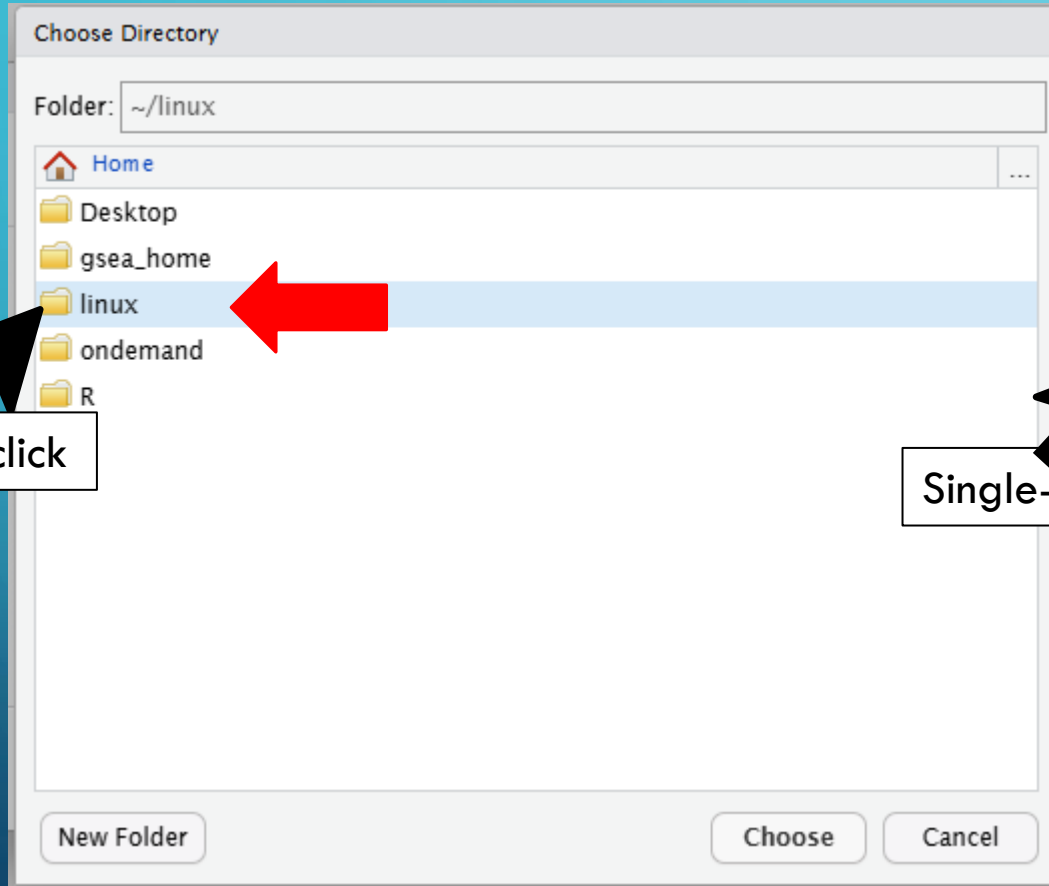
CREATING A PROJECT



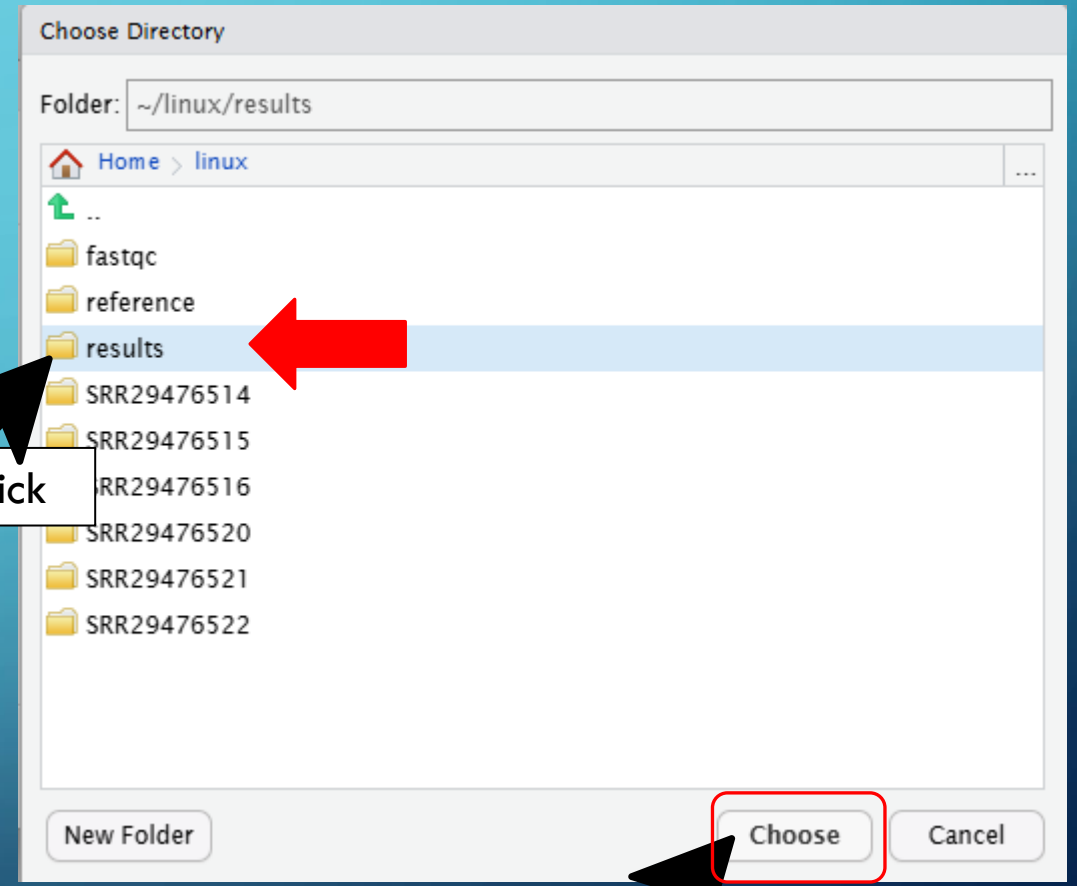
CREATING A PROJECT



CREATING A PROJECT



Double-click



Single-click

CREATING A PROJECT

New Project Wizard

[Back](#) **Create New Project**



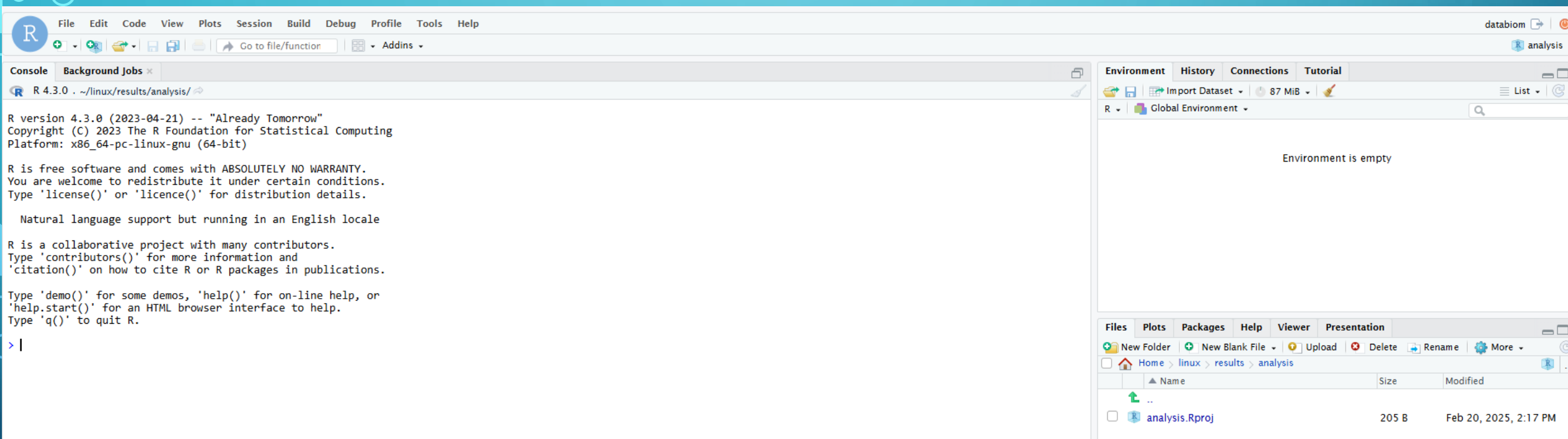
Directory name:

Create project as subdirectory of:
 [Browse...](#)

☐ Use renv with this project

[Create Project](#) [Cancel](#)

NEW PROJECT



The screenshot displays the RStudio IDE interface for a new project. The top menu bar includes File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. The toolbar contains icons for file operations and a search bar. The console window shows the R version 4.3.0 startup message, including copyright information and usage instructions. The Environment pane is empty, indicating no objects are loaded in the global environment. The Files pane shows the project structure, with a file named 'analysis.Rproj' listed.

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Console Background Jobs

R 4.3.0 . ~/linux/results/analysis/

R version 4.3.0 (2023-04-21) -- "Already Tomorrow"
Copyright (C) 2023 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> |

Environment History Connections Tutorial

Import Dataset 87 MiB

R Global Environment

Environment is empty

Files Plots Packages Help Viewer Presentation

New Folder New Blank File Upload Delete Rename More

Home linux results analysis

	Name	Size	Modified
	..		
	analysis.Rproj	205 B	Feb 20, 2025, 2:17 PM

INSTALL NECESSARY PACKAGES

Console

Terminal ✕

Background Jobs ✕

R 4.3.0 . ~/linux/results/analysis/ ➡

```
> BiocManager::install("tximport")|
```

Press "enter"

```
Update all/some/none? [a/s/n]:
```

a

Press "enter"

INSTALL PACKAGES

Console

Terminal ✕

Background Jobs ✕

 R 4.3.0 . ~/linux/results/analysis/ ➡

```
> BiocManager::install("apeglm")
```

Press "enter"

INSTALL PACKAGES

Console

Terminal ✕

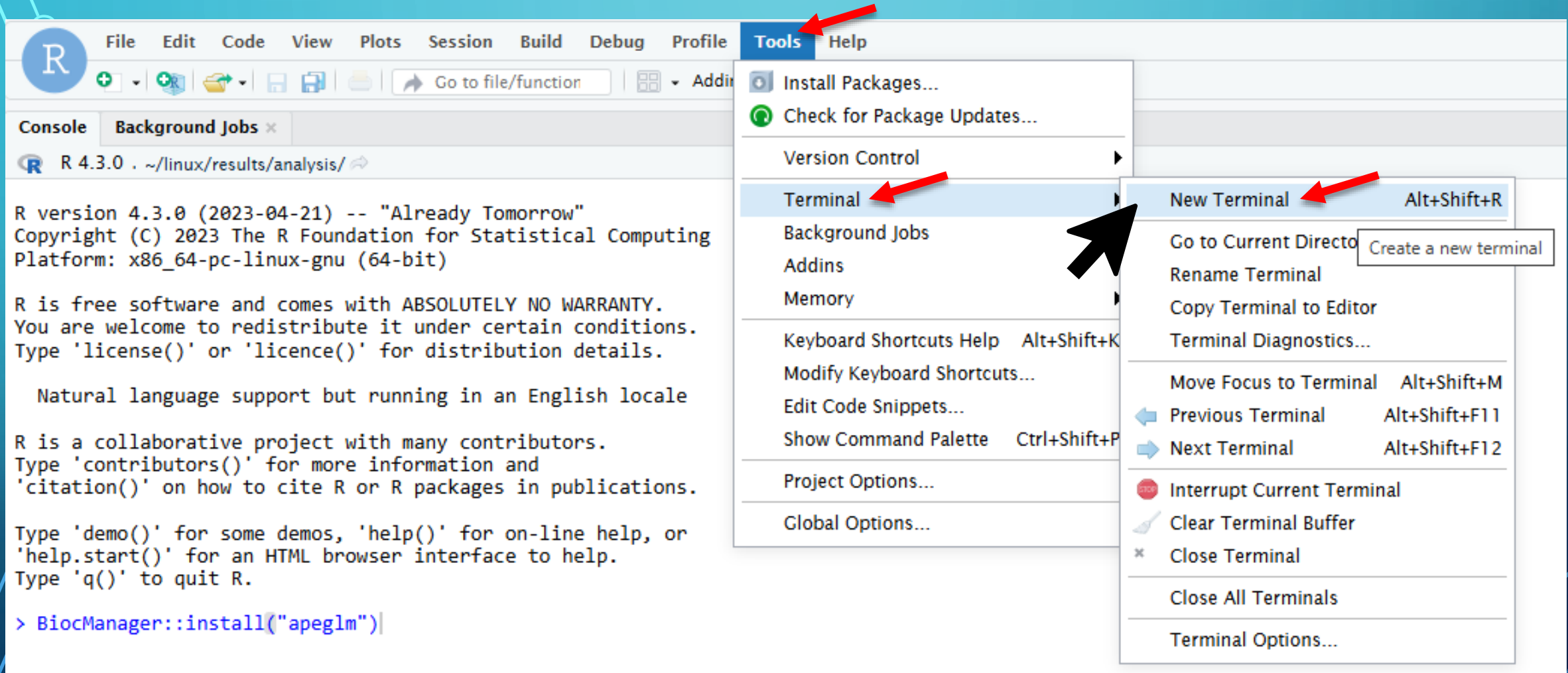
Background Jobs ✕

 R 4.3.0 . ~/linux/results/analysis/ 

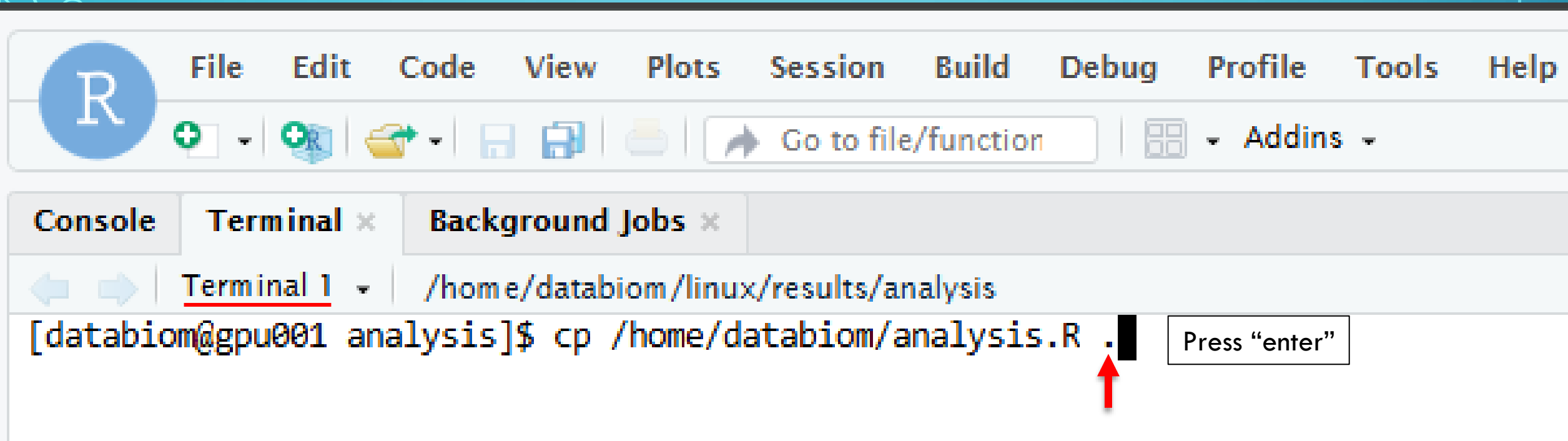
```
> BiocManager::install("org.Hs.eg.db")
```

Press "enter"

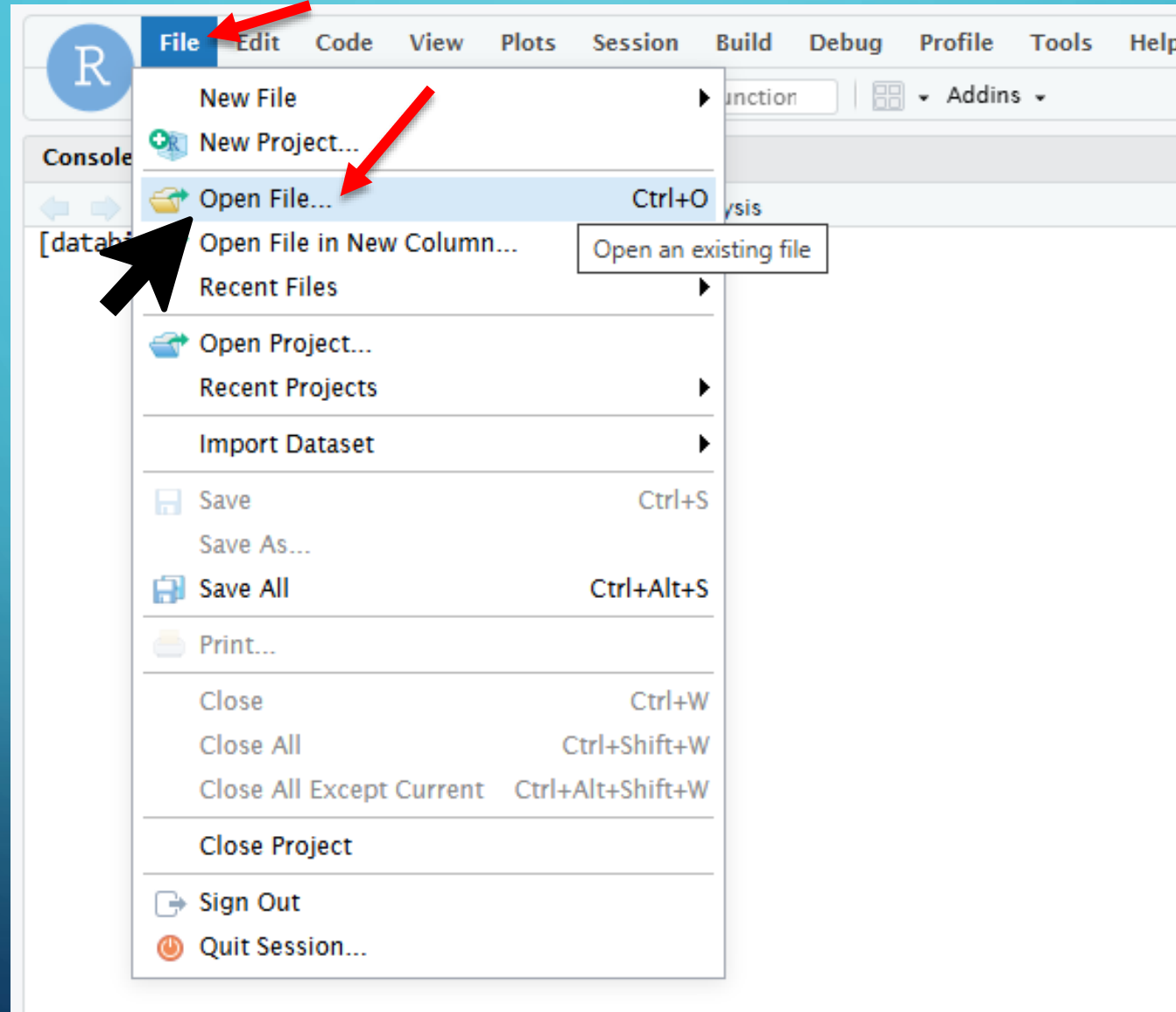
OPEN A TERMINAL WITHIN RSTUDIO



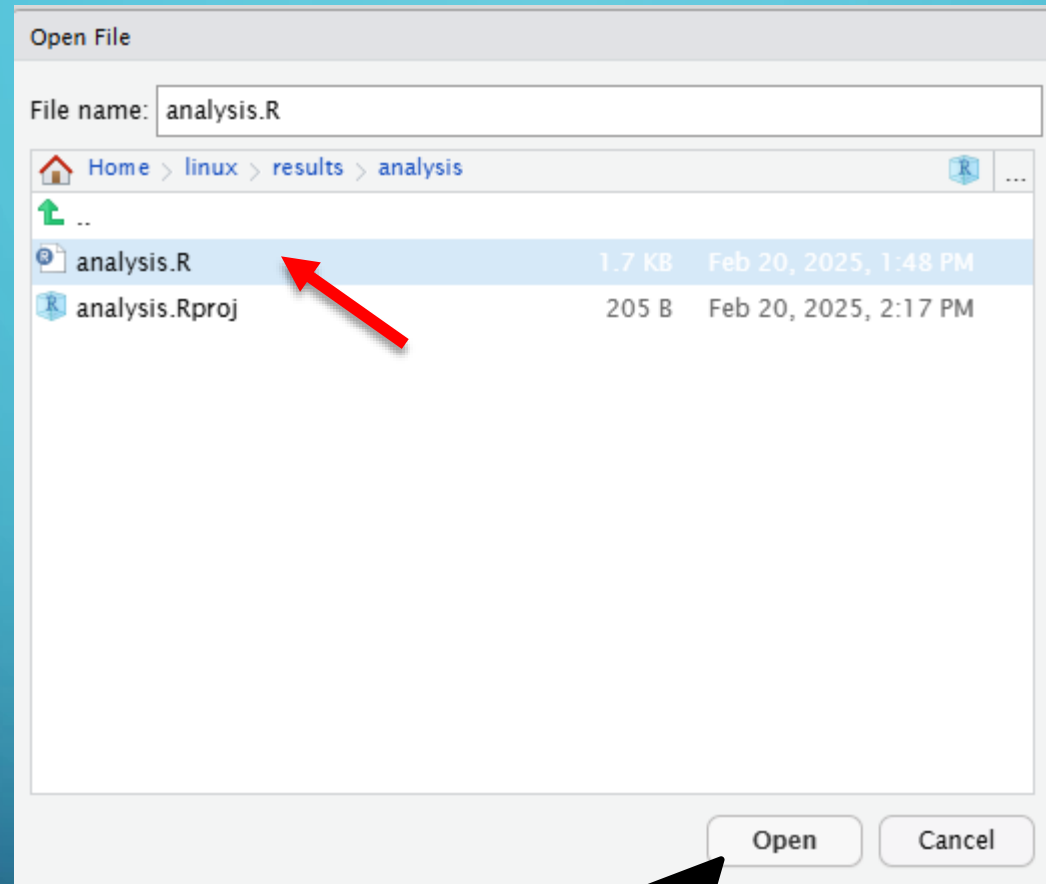
COPY AN R SCRIPT TO YOUR DIRECTORY



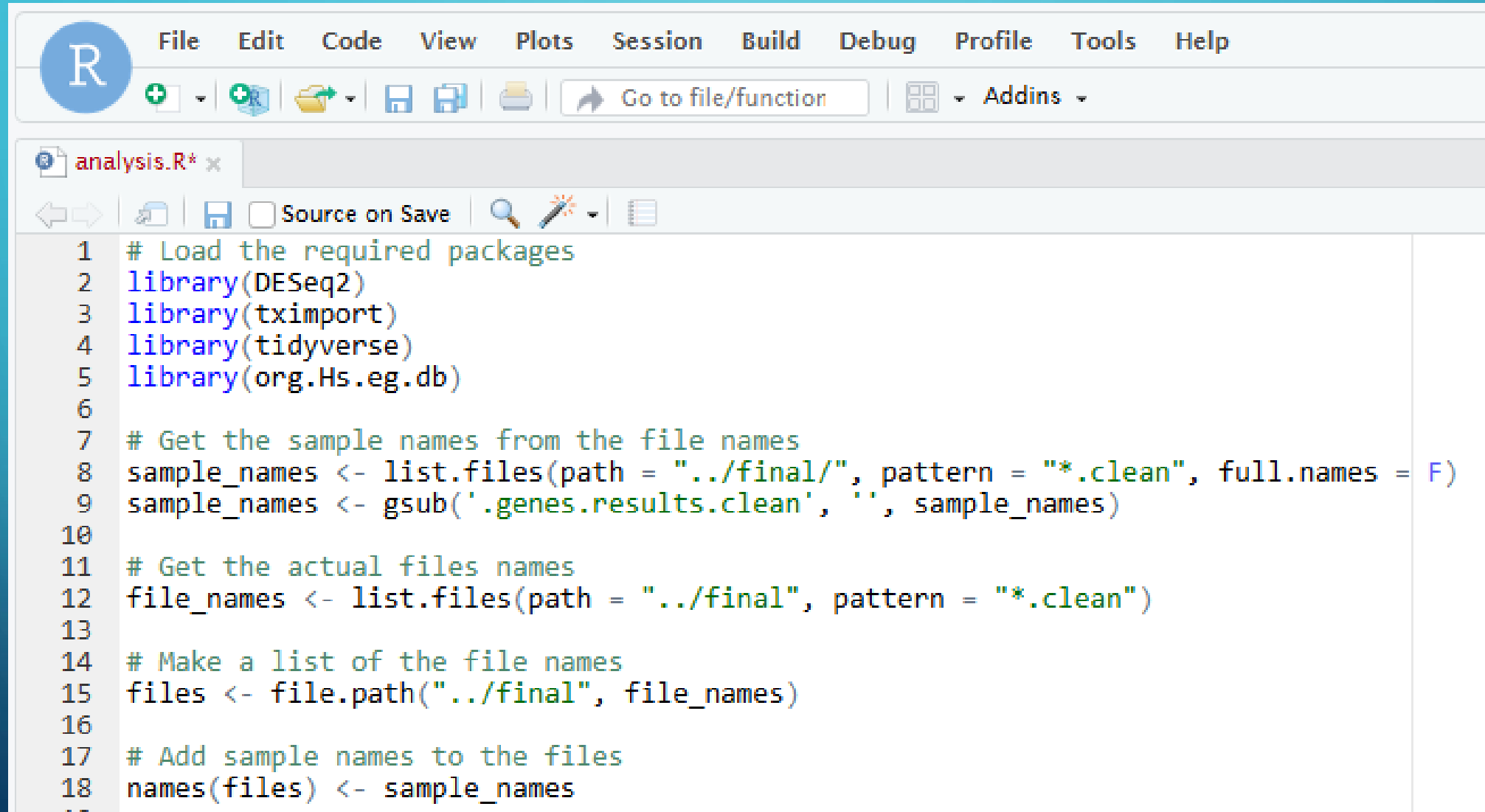
OPEN THE SCRIPT



OPEN THE SCRIPT

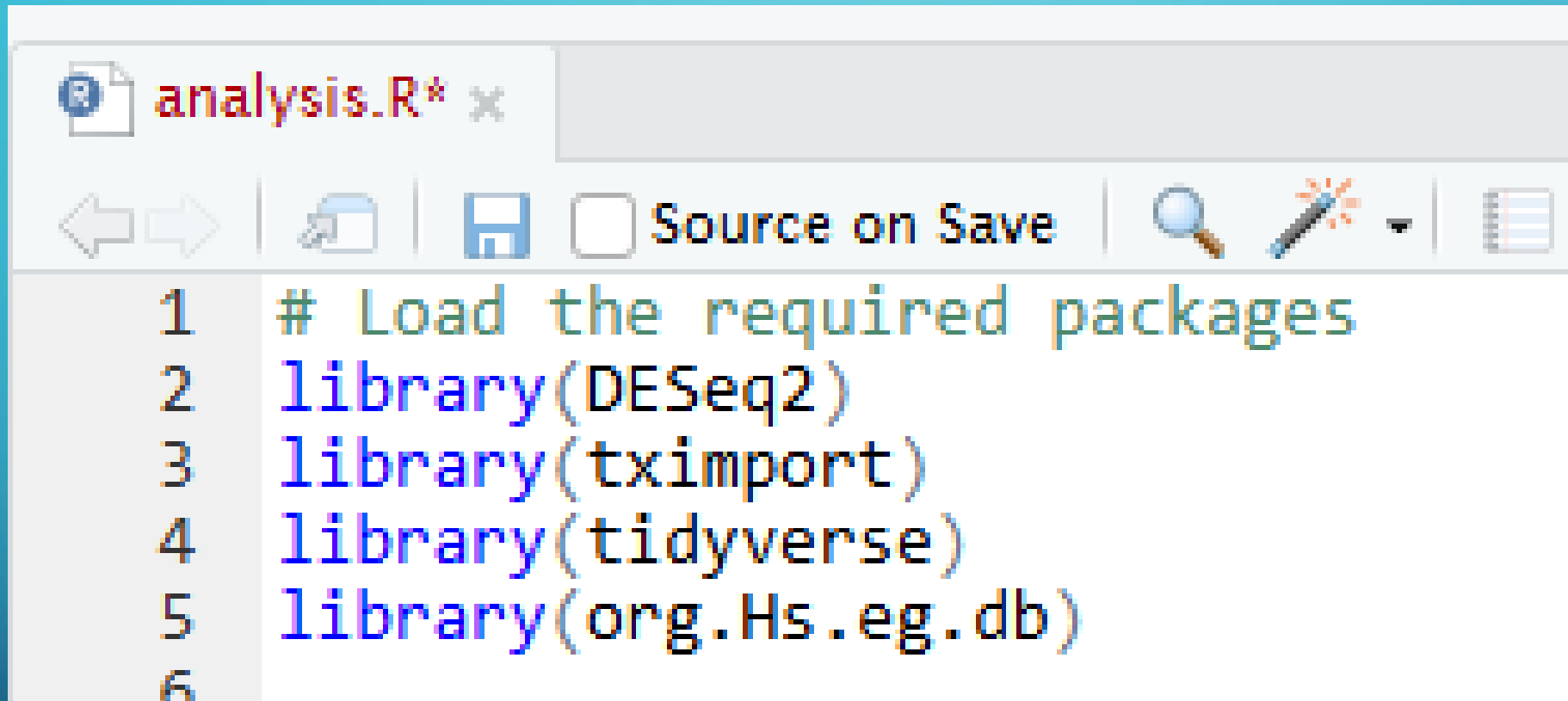


SCRIPT



```
1 # Load the required packages
2 library(DESeq2)
3 library(tximport)
4 library(tidyverse)
5 library(org.Hs.eg.db)
6
7 # Get the sample names from the file names
8 sample_names <- list.files(path = "../final/", pattern = "*.clean", full.names = F)
9 sample_names <- gsub('.genes.results.clean', '', sample_names)
10
11 # Get the actual files names
12 file_names <- list.files(path = "../final", pattern = "*.clean")
13
14 # Make a list of the file names
15 files <- file.path("../final", file_names)
16
17 # Add sample names to the files
18 names(files) <- sample_names
```

LOAD REQUIRED PACKAGES



The screenshot shows an R script editor window with the title 'analysis.R*'. The script contains the following lines:

```
1 # Load the required packages
2 library(DESeq2)
3 library(tximport)
4 library(tidyverse)
5 library(org.Hs.eg.db)
6
```

Use ctrl+Enter to execute each line of the script one by one

If you don't have DESeq2 or Tidyverse installed, you need to install them first:

```
BiocManager::install("DESeq2")
install.packages("tidyverse")
```

MAKE VARIABLES FOR FILE AND SAMPLE NAMES

```
7 # Get the sample names from the file names
8 sample_names <- list.files(path = "../final/", pattern = "*.clean", full.names = F)
9 sample_names <- gsub('.genes.results.clean', '', sample_names)
10
11 # Get the actual files names
12 file_names <- list.files(path = "../final", pattern = "*.clean")
13
14 # Make a list of the file names
15 files <- file.path("../final", file_names)
16
17 # Add sample names to the files
18 names(files) <- sample_names
```

IMPORT FILES

```
20 # Use tximport to import the files and create a matrix
21 txi <- tximport(files, type = "rsem", txIn = FALSE, txOut = FALSE)
22
23 # Change genes with length zero to 0.01
24 txi$length[txi$length == 0] <- 0.01
25
26 # check matrix
27 head(txi$counts)
```

COUNT MATRIX

```
> head(txi$counts)
```

	SRR29476514	SRR29476515	SRR29476516	SRR29476520	SRR29476521	SRR29476522
ENSG000000000003	674.00	663.00	624.00	756.00	676.00	437.00
ENSG000000000005	0.00	0.00	0.00	0.00	0.00	0.00
ENSG0000000000419	1429.72	1500.27	1275.08	1584.06	1464.55	946.25
ENSG0000000000457	271.69	328.20	323.40	241.94	284.75	217.81
ENSG0000000000460	345.31	440.80	368.60	385.06	492.25	382.19
ENSG0000000000938	1.00	0.00	0.00	0.00	0.00	0.00

```
> |
```

MAKE A DATAFRAME FOR OUR TREATMENTS

```
30 # Create a dataframe for sample names and treatments
31 treatment <- c("over", "over", "over", "control", "control", "control")
32 colData <- data.frame(treatment, row.names = sample_names)
33
34 # Check our dataframe
35 head(colData)
```

Console output:

```
> head(colData)
      treatment
SRR29476514   over
SRR29476515   over
SRR29476516   over
SRR29476520 control
SRR29476521 control
SRR29476522 control
> |
```

MAKE SURE MATRIX AND DATAFRAME AGREE

```
37 # Check that the matrix and dataframe have the same information
38 all(rownames(colData) %in% colnames(txi$counts))
39 all(rownames(colData) == colnames(txi$counts))
40 which(rownames(colData) != colnames(txi$counts))
```

Console output:

```
> # Check that the matrix and dataframe have the same information
> all(rownames(colData) %in% colnames(txi$counts))
[1] TRUE
> all(rownames(colData) == colnames(txi$counts))
[1] TRUE
> which(rownames(colData) != colnames(txi$counts))
integer(0)
> |
```

CREATE DESEQ DATASET

```
42 # Create a DESeq dataset
43 dds <- DESeqDataSetFromTximport(txi, colData, ~treatment)
```

Console output:

```
> dds <- DESeqDataSetFromTximport(txi, colData, ~treatment)
using counts and average transcript lengths from tximport
Warning message:
In DESeqDataSet(se, design = design, ignoreRank) :
  some variables in design formula are characters, converting to factors
> |
```

FILTER OUT LOWLY EXPRESSED GENES

```
45 # Check how many rows (genes)
46 nrow(dds) #78,932
47
48 # Filter out low counts ( keep only rows that have a count of at least 10 for a minimal number of samples)
49 smallestGroupSize <- 3
50 keep <- rowSums(counts(dds) >= 10) >= smallestGroupSize
51 dds <- dds[keep,]
52
53 # Check number of genes remaining
54 nrow(dds) #16,612
```

Console output:

```
> # Check how many rows (genes)
> nrow(dds) #78,932
[1] 78932
> # Filter out low counts ( keep only rows that have a count of at least 10 for a minimal number of samples)
> smallestGroupSize <- 3
> keep <- rowSums(counts(dds) >= 10) >= smallestGroupSize
> dds <- dds[keep,]
> # Check number of genes remaining
> nrow(dds) #16,612
[1] 16612
>
```

SET CONTROL AND RUN DESEQ

```
56 # Set control
57 dds$treatment <- relevel(dds$treatment, ref = "control")
58
59 # Run DESeq
60 dds <- DESeq(dds)
61 res <- results(dds)|
```

```
> # Set control
> dds$treatment <- relevel(dds$treatment, ref = "control")
> # Run DESeq
> dds <- DESeq(dds)
estimating size factors
using 'avgTxLength' from assays(dds), correcting for library size
estimating dispersions
gene-wise dispersion estimates
mean-dispersion relationship
final dispersion estimates
fitting model and testing
```

ADD GENE SYMBOLS

```
63 # Add gene symbols to our results
64 res$symbol <- mapIds(org.Hs.eg.db, keys = row.names(res),
65                      column = "SYMBOL",
66                      keytype = "ENSEMBL",
67                      multiVals = "first")
```

```
> res$symbol <- mapIds(org.Hs.eg.db, keys = row.names(res),
+                      column = "SYMBOL",
+                      keytype = "ENSEMBL",
+                      multiVals = "first")
'select()' returned 1:many mapping between keys and columns
```

SUMMARY OF RESULTS

```
69 # Get a summary of the results
70 summary(res)
```

```
> # Get a summary of the results
> summary(res)

out of 16612 with nonzero total read count
adjusted p-value < 0.1
LFC > 0 (up)           : 242, 1.5%
LFC < 0 (down)         : 74, 0.45%
outliers [1]           : 32, 0.19%
low counts [2]         : 1289, 7.8%
(mean count < 14)
[1] see 'cooksCutoff' argument of ?results
[2] see 'independentFiltering' argument of ?results
```

ORDER RESULTS BY STATISTICAL SIGNIFICANCE

```
72 # reorder the results by p-value (most statistically significant)
73 resOrdered <- res[order(res$pvalue),]
74
75 # Most significant gene is ENSG00000118217 (ATF6), which is the gene that is overexpressed in this experiment
76 head(resOrdered)
```

```
> head(resOrdered)
log2 fold change (MLE): treatment over vs control
Wald test p-value: treatment over vs control
DataFrame with 6 rows and 7 columns
```

	baseMean	log2FoldChange	lfcSE	stat	pvalue	padj	symbol
	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>	<numeric>	<character>
ENSG00000118217	17939.389	5.23252	0.131211	39.87860	0.00000e+00	0.00000e+00	ATF6
ENSG00000150961	1221.688	1.91682	0.195228	9.81833	9.38889e-23	7.17828e-19	SEC24D
ENSG0000013375	768.558	1.18987	0.130416	9.12370	7.26038e-20	3.70061e-16	PGM3
ENSG00000155660	12808.488	1.11321	0.124618	8.93301	4.14572e-19	1.58481e-15	PDIA4
ENSG00000051108	1797.897	1.96031	0.226695	8.64734	5.27142e-18	1.61211e-14	HERPUD1
ENSG00000102401	916.880	1.37204	0.159103	8.62356	6.49020e-18	1.61377e-14	ARMCX3

```
> |
```

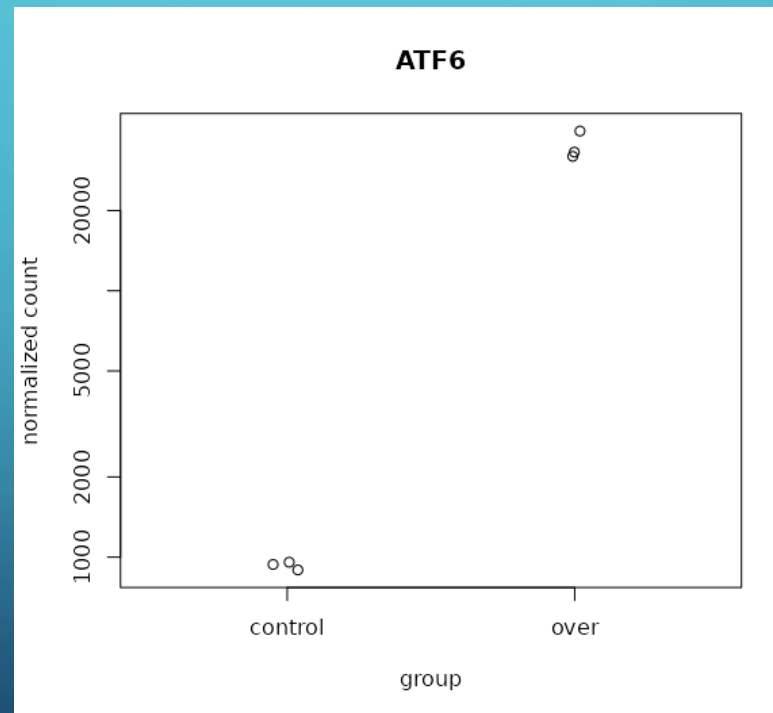


REDUCE INFLATED LOG_2 FOLD CHANGE VALUES

```
78 # Shrinkage of effect size (LFC estimates) is useful for visualization and ranking of genes
79 resLFC <- lfcShrink(dds,coef = "treatment_over_vs_control", type = "apeglm")
```

PLOT TRANSCRIPT COUNTS OF A GENE

```
81 # Use the plotCounts function to plot the counts for ATF6 for both treatments
82 plotCounts(dds, gene=which.min(res$padj), intgroup="treatment", main = "ATF6")
83
```



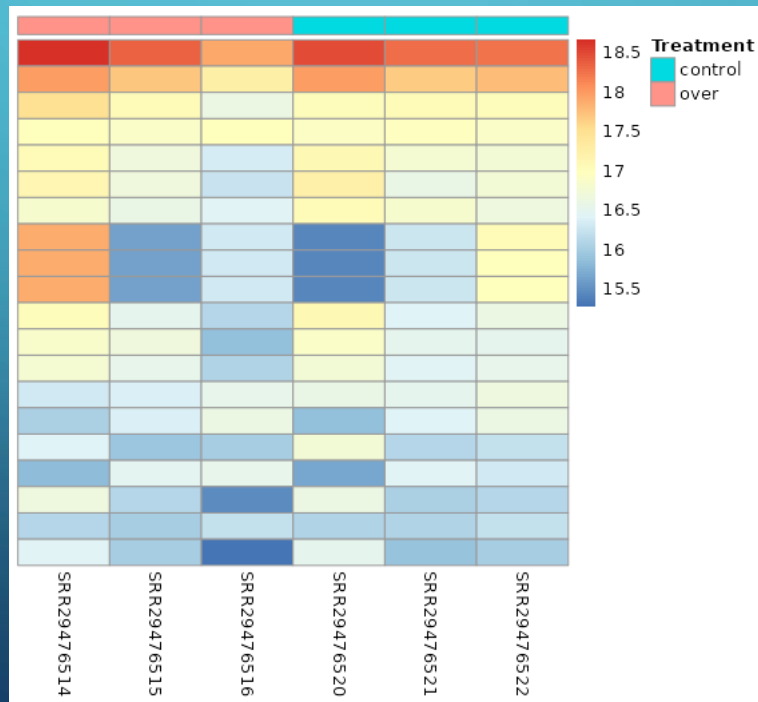
TRANSFORM DATA AND INSTALL PHEATMAP PACKAGE

```
84 # Use the variance stabilizing transformation (vst) function (good for visualization or clustering)
85 vsd <- vst(dds, blind=FALSE)
```

```
87 # Install and load the heatmap package "pheatmap"
88 BiocManager::install("pheatmap")
89 library(pheatmap)|
```

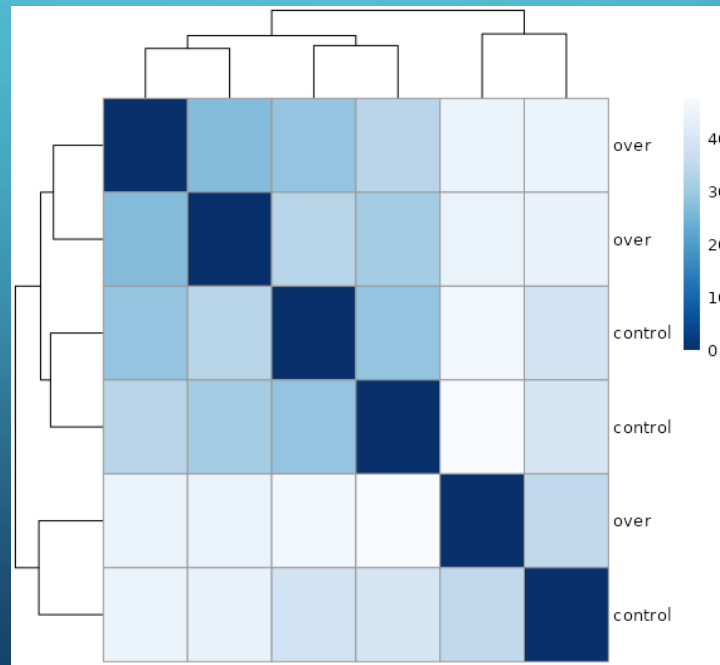
HEATMAP

```
91 # Create a heatmap of the top 20 highest normalized counts
92 select <- order(rowMeans(counts(dds,normalized=TRUE)),
93                 decreasing=TRUE)[1:20]
94 df <- as.data.frame(colData(dds)[,"treatment"])
95 rownames(df) <- colnames(dds)
96 colnames(df) = "Treatment"
97 pheatmap(assay(vsd)[select,], cluster_rows=FALSE, show_rownames=FALSE,
98          cluster_cols=FALSE, annotation_col=df, annotation_names_col=FALSE)
```



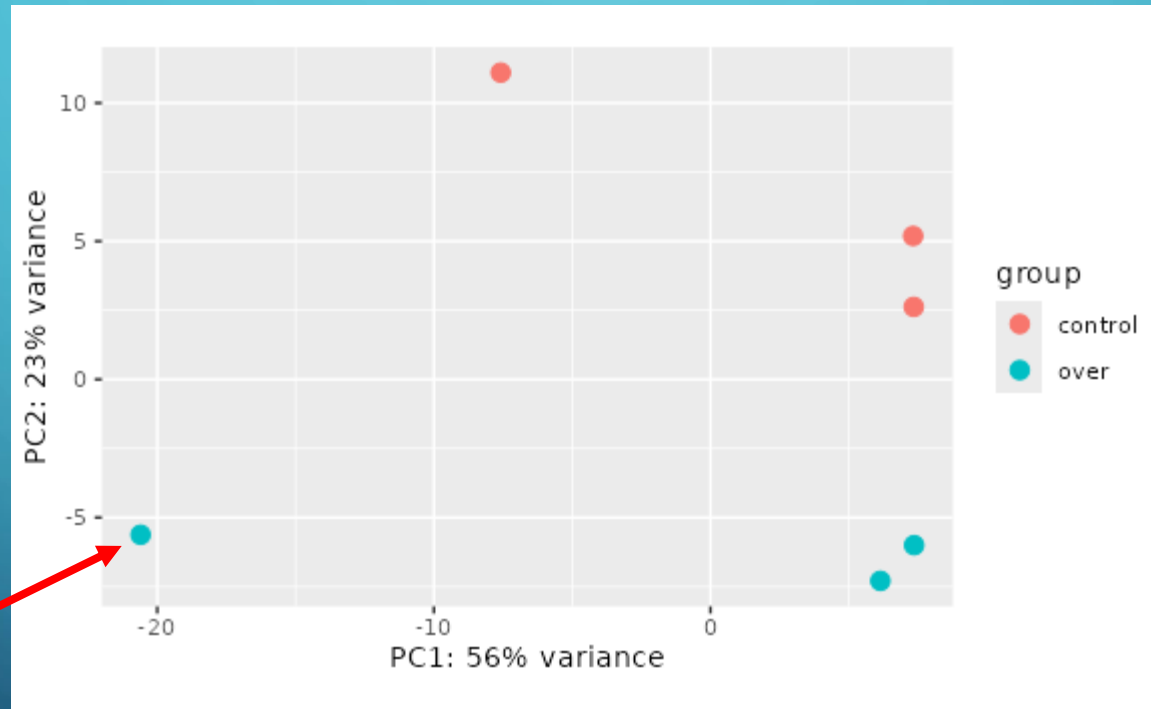
DISTANCE MATRIX

```
100 # Create a heatmap of the distance matrix to see similarities and dissimilarities between samples
101 library("RColorBrewer")
102 sampleDists <- dist(t(assay(vsd)))
103 sampleDistMatrix <- as.matrix(sampleDists)
104 rownames(sampleDistMatrix) <- paste(vsd$treatment)
105 colnames(sampleDistMatrix) <- NULL
106 colors <- colorRampPalette( rev(brewer.pal(9, "Blues")) )(255)
107 pheatmap(sampleDistMatrix,
108           clustering_distance_rows=sampleDists,
109           clustering_distance_cols=sampleDists,
110           col=colors)
```



PRINCIPAL COMPONENT ANALYSIS (PCA)

```
112 #Create a principal component analysis (PCA) plot
113 # useful for visualizing the overall effect of experimental covariates and batch effects
114 plotPCA(vsd, intgroup="treatment")
```



GOOD JOB!

We'll go over some more advanced visualization methods and pathway analyses next time.

Don't forget to save your script if you made any edits.

